Extraordinary transparency in continuous metallic thin film based on transition metal dichalcogenides metasurfaces

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Two dimensional materials have emerged as a toolbox for nanophotonics and nano-optoelectronics providing new properties that are not possible in conventional materials. In particular semiconductor transition metal dichalcogenides (TMDC) show unique optical properties. In this presentation we propose a nanostructure based on TMDC which achieves extraordinary transmission through a continuous thin metallic film.

So, we have designed a metasurface of TMDC on top and under the continuous metal film, obtaining a final geometry TMDC-metal-TMDC (Fig.1a). Due to the anisotropic high refractive index of the TMDC metasurface photonic plasmonic resonances are created (Fig. 1b) obtaining a high transmission through the continuous metallic film. By means of FDTD simulations we find configurations showing up to 90% transmission for a 30 nm thick metal film of silver, as shown in Figure 1c, which is an enhancement 10 times higher than the transmission of the metal without nanostructure.

This high transmission is tunable by means of the variation of the geometric parameter of the metasurface. This effect could be suitable for actual applications in optoelectronic devices, such as transparent solar cell electrodes, plasmonic filters, polarizers or to increase the efficiency of resonant cavities for photodetectors.

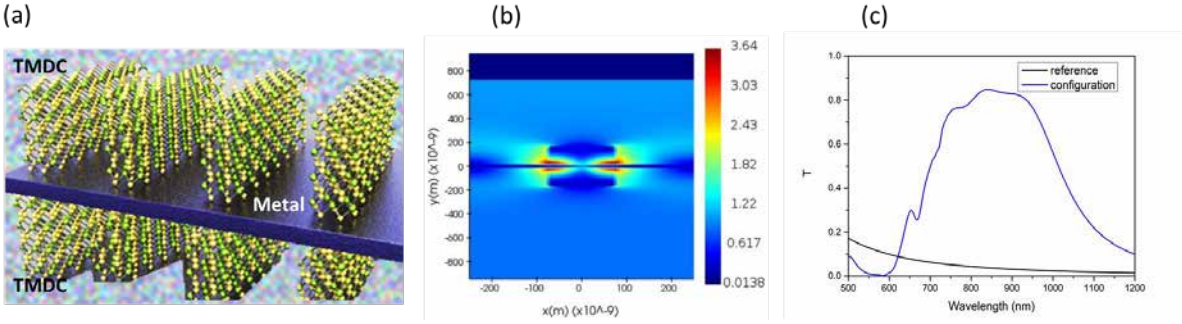


***Figure 1**. (a) Design of the nanostructure based on TMDC-metal-TMDC. (b) Plasmon resonance originated in the photonic architecture. (c)* Transmission with one nanostructure based on MoS2-metal-MoS2 (blue line) in comparison with bare metal transmission (black line).

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References:

[1]Y. Wakabayashi, T. Fukui, J. Yamauchi and H. Nakano, 2012 International Symposium on Antennas and Propagation (ISAP), 2012, pp. 648-651.